

## Study Overview

### Brief Summary

Fusion of anaplastic lymphoma kinase (ALK) is an important driving gene for NSCLC, with an incidence rate of 3-7%. In patients with advanced ALK mutation NSCLC, first-line use of ALK inhibitors significantly improves progression free survival. The perioperative research on ALK positive NSCLC was relatively late, and currently most studies mainly focus on early to mid stage ALK positive NSCLC patients. The results of two Phase III clinical trials showed that second-generation ALK targeted drugs, neoadjuvant and/or adjuvant therapy for ALK positive NSCLC, significantly prolonged DFS in patients, including increased pathological response rate, median response duration, and prolonged OS. For ALK positive advanced NSCLC patients who are resistant to second-generation ALK targeted drugs, there is currently limited exploration and there is an urgent need for new exploratory clinical studies. This trial aims to evaluate the effectiveness of Iruplinalkib neoadjuvant therapy for potentially resectable ALK positive non-small cell lung cancer.

### Official Title

Neoadjuvant Therapy With Iruplinalkib for Potentially Resectable ALK Positive Non-Small Cell Lung Cancer: A Single Arm, Exploratory Trial

### Contacts and Locations

This section provides contact details for people who can answer questions about joining this study, and information on where this study is taking place.

To learn more, please see the Contacts and Locations section in How to Read a Study Record.

### Study Contact

Name: Peng Zhang, PhD

Phone Number: 021-65115006

Email: zhangpeng1121@tongji.edu.cn

### Study Contact Backup

Name: Shuangyi Li, PhD

Email: dr.frederic.lee@gmail.com

### Participation Criteria

Researchers look for people who fit a certain description, called eligibility criteria. Some examples of these criteria are a person's general health condition or prior treatments.

For general information about clinical research, read Learn About Studies.

### Eligibility Criteria

#### Description

#### Inclusion Criteria:

1. The patient shall sign the Informed Consent Form.
2. Aged  $18 \geq$  years.
3. Histological or cytological diagnosis of NSCLC by needle biopsy, and evaluated by researchers as stage III-IVA, and diagnosed as ALK positive through genetic testing.
4. Eastern Cooperative Oncology Group (ECOG) performance-status score of 0 or 1.
5. According to the MDT evaluation (which should include a thoracic surgeon specializing in tumor surgery), it is considered that the primary NSCLC is potentially completely resectable;
6. At least 1 measurable lesion according to RECIST 1.1.
7. Patients with good function of other main organs (liver, kidney, blood system, etc.)
8. Patients with lung function can tolerate surgery;
9. Fertile female patients must voluntarily use effective contraceptives not less than 120 days after chemotherapy or the last dose of toripalimab (whichever is later) during the study period, and urine or serum pregnancy test results within 7 days prior to enrollment are negative.
10. Unsterilized male patients must

voluntarily use effective contraception during the study period not less than 120 d

#### Exclusion Criteria:

1. Previously received targeted therapy (including TKI or monoclonal antibodies), immunotherapy, or any investigational drug treatment for NSCLC;
2. Pathological confirmation of mixed small cell and non-small cell lung cancer;
3. Patients with a malignancy other than NSCLC within five years prior to the start of this trial, except for cured basal cell carcinoma of the skin, early gastrointestinal (GI) carcinoma excised through endoscopy, cervical carcinoma in situ, ductal carcinoma in situ of the breast, papillary thyroid carcinoma, or any cured cancer deemed to have no impact on the survival of the current NSCLC;
4. Participants with any unstable systemic disease (including active infection, uncontrolled hypertension), unstable angina pectoris, angina pectoris starting in the last three months, congestive heart failure ( $\geq$  NYHA Grade II), myocardial infarction (6 months before admission), severe arrhythmia requiring drug treatment, liver, kidney or metabolic diseases;
5. Patients with congenital or acquired immune dysfunction (such as HIV infected individuals);
6. Received other major surgical treatments (excluding diagnosis) within 4 weeks prior to the start of the study or expected to undergo major surgical treatments during the study period;
7. Participants who are allergic to the test drug or any auxiliary materials;
8. A history of extensive diffuse bilateral interstitial fibrosis in the past or before medication, or a known grade 3 or 4 history of interstitial fibrosis or interstitial lung disease, including pneumonia, allergic pneumonia, interstitial pneumonia, interstitial lung disease, obliterative bronchiolitis, and pulmonary fibrosis, but not including local radiation pneumonia or radiation pulmonary fibrosis history;
8. Pregnant or lactating women;
9. Any malabsorption;
10. Participants suffering from nervous system diseases or mental diseases that cannot cooperate
11. Other factors that researchers think it is not suitable for enrollment.

#### Study Plan

This section provides details of the study plan, including how the study is designed and what the study is measuring.

Expand all / Collapse all

How is the study designed?

Design Details

Primary Purpose : Treatment

Allocation : N/A

Interventional Model : Single Group Assignment

Interventional Model Description: All enrolled patients received Iruplinalkib neoadjuvant therapy.

Masking : None (Open Label)

Arms and Interventions

Participant Group/Arm Intervention/Treatment

Experimental: Neoadjuvant Therapy With Iruplinalkib

In this arm, 20 patients with III-IVA stage, potentially resectable ALK Positive non-small cell lung cancer will receive 4 circles of neoadjuvant Iruplinalkib. And the patients who are resectable after neoadjuvant therapy will be treated with surgery. Postoperative patients receive adjuvant Iruplinalkib therapy up to 2 years/until the disease progressed.

Drug: Iruplinalkib

60 mg on days 1-7, if tolerable, 180 mg from day 8 onwards.

Procedure: surgery

Patients with resectable tumors after neoadjuvant therapy will be treated with surgery.

What is the study measuring?

Primary Outcome Measures

| Outcome Measure | Measure Description | Time Frame |
|-----------------|---------------------|------------|
|-----------------|---------------------|------------|

Objective response rate (ORR) ORR is defined according to the RECIST v1.1 criteria. up to 30 months

#### Secondary Outcome Measures

| Outcome Measure | Measure Description | Time Frame |
|-----------------|---------------------|------------|
|-----------------|---------------------|------------|

|                                         |                                                                                                                                                                                                                              |                 |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Pathologic complete response (PCR) rate | PCR rate is defined as the proportion of participants who have achieved pathologic complete response (on routine hematoxylin and eosin staining, no tumor cell can be found in tumor bed or lymph node) in all participants. | Up to 30 months |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|

|                                      |                                                                                                                                                                                                                     |                 |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Major pathologic response (MPR) rate | MPR rate is defined as the proportion of participants who have achieved major pathologic response (on routine hematoxylin and eosin staining, tumors with no more than 10% viable tumor cells) in all participants. | Up to 30 months |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|

|                       |                                                                                                                                                                                                                                 |                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Overall survival (OS) | It is defined as the time (months) from enrollment to death of participant due to any cause. In the case of a patient who still survives at the time of analysis, the date of last contact will be taken as the censoring date. | up to 60 months |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Event-free survival (EFS) | Event-free survival (EFS) is defined as the length of time (months) from randomization to any of the following events: any progression of disease precluding surgery, progression or recurrence disease based on response evaluation criteria in solid tumors (RECIST) 1.1 after surgery, or death due to any cause. Participants who don't undergo surgery for reason other than progression will be considered to have an event at progression or death. Progression is defined as at least a 20% increase in the sum of diameters of target lesions, taking as reference the smallest sum on study. In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm. (Note: the appearance of one or more new lesions is also considered progression). | up to 60 months |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|

|                                        |                                                                                 |                 |
|----------------------------------------|---------------------------------------------------------------------------------|-----------------|
| Treatment-related adverse event (TRAE) | TRAE is defined and classified according to NCI-CTCAE v5.0 in all participants. | Up to 30 months |
|----------------------------------------|---------------------------------------------------------------------------------|-----------------|

|                              |                                                                                                                                                                                                                                 |                 |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 5-year overall survival (OS) | OS is defined as the time (months) from enrollment to death of participant due to any cause. In the case of a patient who still survives at the time of analysis, the date of last contact will be taken as the censoring date. | Up to 72 months |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|

#### 1-year event-free survival (EFS) rate

Event-free survival (EFS) is defined as the length of time (months) from randomization to any of the following events: any progression of disease precluding surgery, progression or recurrence disease based on response evaluation criteria in solid tumors (RECIST) 1.1 after surgery, or death due to any cause. Participants who don't undergo surgery for reason other than progression will be considered to have an event at progression or death. Progression is defined as at least a 20% increase in the sum of diameters of target lesions, taking as reference the smallest sum on study. In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm. (Note:

the appearance of one or more new lesions is also considered progression).

Up to 24 months

Health related quality of life (HRQoL) The assessment is made according to the Quality of Life Scale for Lung Cancer Patients (EORTC-QLQ-C30 & LC13, Version 3). EORTC's QLQ-C30 & LC13 (V3.0) is a core scale for lung cancer patients, with a total of 43 items. Among them, Item 29 and 30 are divided into seven grades, which are assigned with 1 to 7 scores according to the answer options. The other items are divided into 4 grades: Not at All, A Little, Quite a Bit, and Very Much, assigned with 1 to 4 scores respectively. The higher score, the worse quality. Up to 36 months

#### Collaborators and Investigators

This is where you will find people and organizations involved with this study.

#### Sponsor

Shanghai Pulmonary Hospital, Shanghai, China